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Palaeomagnetism of a Vertical Sequence of Traps from Mount Girnar, Gujrat, India

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Summary

Palaeomagnetic studies on 18 volcanic flows belonging to Deccan trap activity and ranging in elevation from 200 to 2470 ft from Mount Girnar located in the southern part of Kathiawar peninsula of Gujarat are reported. The flows between 200 and 360 ft elevation were found to have reverse polarity while those above were found to be normal. The field reversal observed here is considered to be the same as that observed at other places over the Deccan trap territory at different elevations. Virtual Geomagnetic Poles corresponding to mean direction for each flow are reported. The results are interpreted, in terms of palaeosecular variation and drift of the Indian land mass during the period of Deccan trap activity.

1. Introduction

The Girnar hills in Kathiawar peninsula of Gujarat, India form a part of the Deccan trap formation of peninsular India. This impressive formation is very similar to the Columbia basalts in U.S.A. and Sera Geral formation in south America. The traps at present cover approximately an area of 200 000 sq. miles extending from nearly 69·5° to 82° E and 16° to 23° N. The present distribution shows that the traps could have covered originally a much larger area, of the order of 500 000 sq. miles including a large segment which now forms a part of the Arabian Sea to the west of Bombay. Palaeomagnetically this is one of the most widely studied formations of the world. Several investigators have contributed to the study of the Deccan traps. Significant contributions have been made by Clegg, Deutsch & Griffiths (1956), Sahsrabudhe (1963), Bhimashankaram & Pal (1968), McElhinny (1968), Pal (1969), Athavale (1970) and Verma & Pullaiah (1972). Fig. 1 shows the location of sites from which Deccan traps have been studied so far.

The early studies by Deutsch, Radhakrishnamurthy & Sahsrabudhe (1959) and Sahsrabudhe (1963) along the west coast of India revealed a pronounced geomagnetic field reversal within the Deccan lavas which was found to be located near an elevation of about 2000 ft above mean sea level. The rocks above this elevation were found to be normally magnetized (i.e. in the present sense of the Earth's field) while those below this elevation were found to be reversely magnetized. Recent studies of Bhimashankaram & Pal (1968) from several localities including Ellora, Aurangabad, Chandor, Ankai, Harsul and Jalana in the north western part of the main Deccan trap exposure and of Verma & Pullaiah (1972) as well as Athavale (1970) from Jabalpur and Amarkantak areas have revealed that this field reversal is located at different

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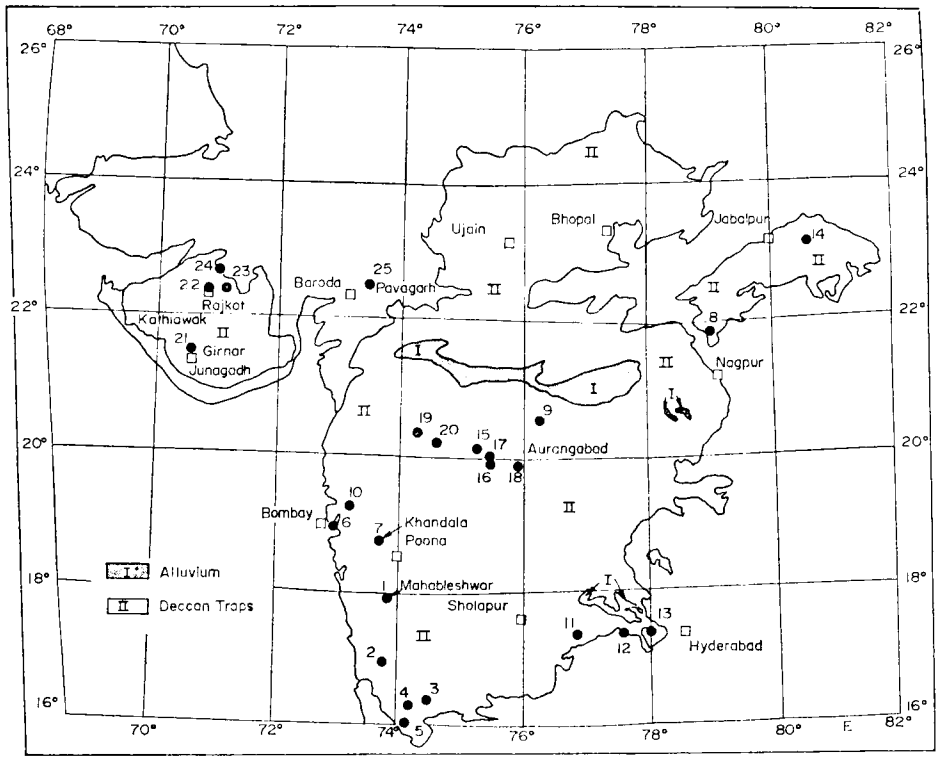


FIG. 1. Map showing the locations of sites from where rocks of Deccan traps have been studied so far. 1. Mahableshwar, 2. Amba, 3. Nipani, 4. Gargoti, 5. Ajra, 6. Bombay, 7. Khandala, 8. Linga, 9. Buldana, 10. Igatpuri, 11. Gulbarga, 12. Chincholi, 13. Vikarabad, 14. Jabalpur, 15. Ellora, 16. Harsul, 17. Aurangabad, 18. Jalna, 19. Chandor, 20. Ankai, 21. Mount Girnar, 22–24. Rajkot and vicinity, 25. Mount Pavagarh.

elevations over most of the territory of the Deccan trap exposure. Bhimashankaram & Pal (1968) have hinted possible existence of more than one reversal within the Deccan trap period. However these have not yet been established with certainty. The data presented so far have indicated only one major field reversal from reverse to normal polarity. In this paper palaeomagnetic study of 18 lava flows from Mount Girnar (which forms a part of the Deccan traps) ranging in elevation from 200 to 2470 ft is reported. The results are interpreted in terms of palaeosecular variation and drift of the Indian Continent during the period of Deccan trap eruptions.

2. Geology of Deccan traps and Mount Girnar

Over most of the territory, the Deccan traps overlie Precambrian crystalline rocks of Peninsular India. In a few areas they also overlie Mesozoic formations. For instance in Jabalpur area the traps overlie Lameta beds, fossil evidence for which suggests the age of Turonian (Middle Cretaceons) according to Krishnan (1960, p. 481). In Narmada valley the traps are underlain by Bagh beds (Cenomian to Senonian age). For some inter-trappean beds, the age of Eocene has been suggested from fossil evidence by Sahni (quoted by Krishnan 1960, p. 484). It has been generally concluded by Oldham as well as Krishnan from geological evidence that the Deccan traps commenced to pour out during Upper Cretaceous times and the activity continued well up to Eocene times (*ibid* p. 485).

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The Girnar hill complex is located in the southern part of the Kathiawar peninsula which is separated from the main Deccan trap exposure by Camaby basin of Tertiary age. Over a large part of the Kutch peninsula Deccan traps are exposed. In the north-eastern part near Wadhwan, Gondwana sandstones called Wadhwan sandstones are underlying the Deccan traps. These are considered to be equivalents of Bagh and Lameta beds (Krishnan 1960, p. 481). The geology of Girnar hill was first studied by Fedden (1884) who described Girnar Hills as a source of volcanic activity towards the close of the Deccan trap period. The Girnar hills represent an igneous complex, the intrusive bodies of which have differentiated into various rock types such as olivine gabbro, diorite, monozonite, microgranite, alkali syenite, nepheline syenite and few other minor rock types. J. W. Evans worked out detailed geology of the Girnar hills. From field study of geology of the area he gave the sequence of of sequence of rocks in the area as follows (quoted from Krishnan 1925).

'The dolerites and basalts of the same characteristics as those of the Deccan traps were extruded first. Immediately following this, gabbros were intruded which are seen to pass into diorites. The next stage of action was alkali-rich syenites. The older rocks mostly diorites and basalts attain prominence in peripheral area while the younger intrusives are confined to the central mass'. Recently Subba Rao (1967) has restudied the area and discussed in detail the sequence of events concerning episodes of igneous activity and petrogenesis of rock types in this igneous complex. He has briefly described the sequence of events in these hills as follows. 'Igneous activity in this region started with the eruption of lavas through the fissures as in the rest of the Deccan traps. Towards the end of fissure types of activity the eruptions of lavas was centralized in the Girnar Hills giving rise to a cone type of volcanic activity. The fissure eruptions gave rise to basic lavas which are almost similar to the typical Deccan trap basalts, while in the central type there is variation in the lavas from the olivine basalts, through oligoclase basalts and andesites to rhyodacites and obsidian due to differentiation of the magma'.

The question of age of the Deccan traps covering the entire span of their eruptions is not yet settled. Recently Kuno (1971) has applied statistical tests to estimate the period of eruption of the Deccan traps taking into account the known field reversals. He remarks, 'application to Deccan basalts show that their volcanic activity could not have lasted more than few million years'.

Wellman & McElhinny (1970) have obtained dates ranging from $59{\cdot}5 \pm 1{\cdot}9$ to $63{\cdot}2 \pm 1{\cdot}2$ My for Mount Pavagarh, $64{\cdot}1 \pm 1{\cdot}9$ My for top flows of Mount Girnar and $59{\cdot}0 \pm 1{\cdot}1$ My for south eastern Deccan traps, 150 km west of Hyderabad. Rama (1968) on the other hand obtained dates ranging from 45 ± 3 My to $60{\cdot}0 \pm 3{\cdot}0$ My for traps from west coast near Bombay. He also obtained ages ranging from 43 ± 2 to 65 ± 5 My for top and bottom flows from Pavagarh. On account of few radiometric ages obtained so far over vast territory of the Deccan traps and unknown factors such as loss of radioactive argon from some of the rocks, the radiometric data obtained so far cannot be considered to be adequate for purposes of estimating the total span of Deccan trap activity. Absence of underlying Mesozoic rocks over large areas covered by the traps adds to this uncertainty.

3. Sampling and laboratory studies

The samples collected for palaeomagnetic studies were obtained from each flow from the base of the hills, which is at an elevation of about 200 ft, up to the top-most flow near an elevation of 3380 ft. At least five to seven hand-oriented samples were collected from each flow. The nature of the flows varied from basalt to diorite, granodiorite and syenite as described by geologists. Elevations of various flows were determined using microaltimeters. These were studied in the palaeomagnetic laboratory of National Geophysical Research Institute at Hyderabad. The sensitivity

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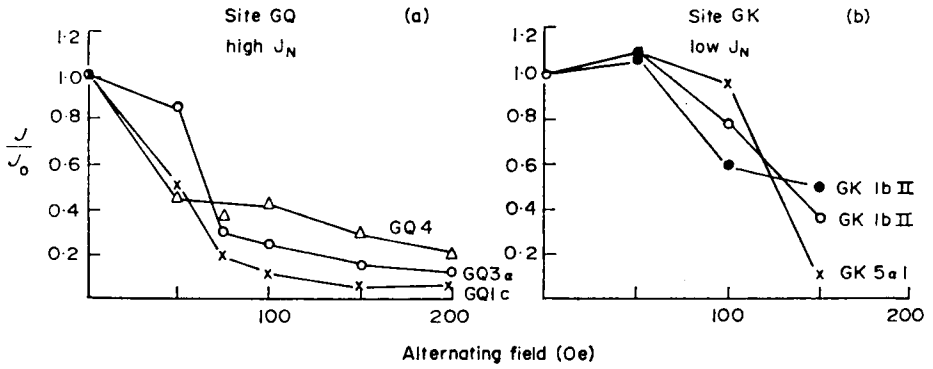


FIG. 2. Typical alternating field demagnetization curves of normalized intensity for rocks possessing high NRM intensity (a), and low NRM intensity (b).

of the astatic magnetometer used was 3×10^{-6} Oe/cm deflection.

Seven to eight pilot specimens from each flow were cleaned in successively increasing alternating fields of 50, 75, 100, 150 and 200 Oe in a demagnetizing apparatus similar to that of Creer (1959). The optimum field necessary for magnetic cleaning for each site was determined and the remaining samples were subjected to demagnetization in that field. Typical a.f. demagnetization curves of normalized intensity for high as well as low NRM intensity (J_N) are shown in Fig. 2.

The samples with high J_N appeared to have acquired appreciable viscous NRM which was appreciably removed in alternating fields of about 75 Oe. The samples with low J_N were found to be more magnetically hard. In NRM measurements the samples from several sites showed both down (+ve) as well as up (−ve) dips. However as a result of a.f. cleaning the directions as well as polarities changed appreciably so that the directions converged towards a site mean direction with either positive or negative dip. Change in direction of r.m. vector with successive alternating fields for one typical specimen is shown in Fig. 3. Fig. 4 shows typical results for four flows (one

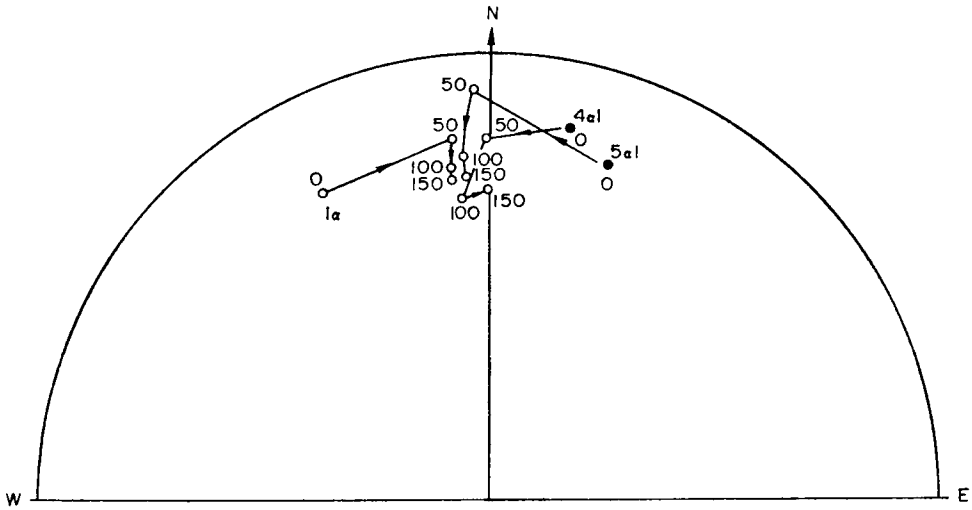


FIG. 3. Change in direction of magnetization as a result of a.f. demagnetization in successively increasing fields, Polar equal area projection is used, ●-down dip, ○-up dip.

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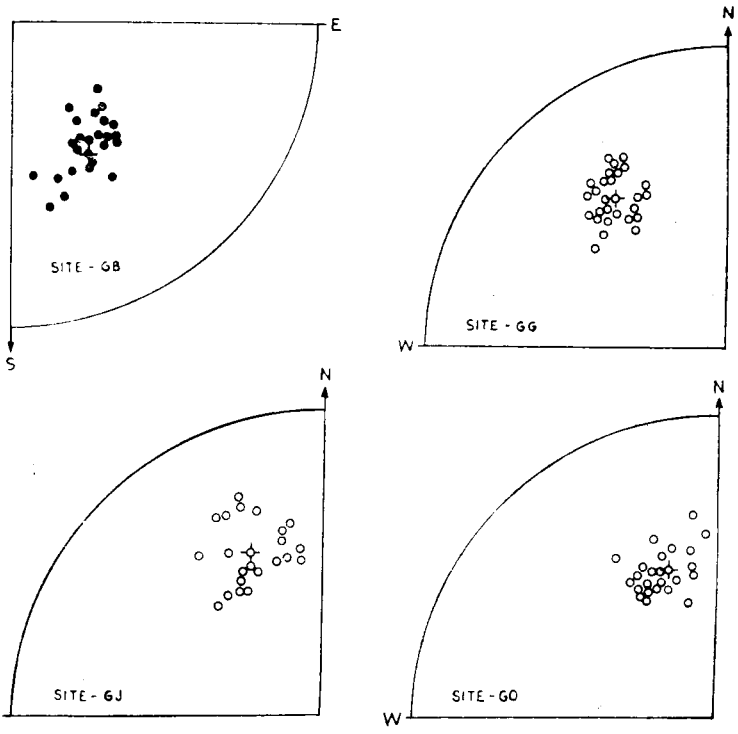


FIG. 4. Directions of magnetization after magnetic cleaning for four typical flows plotted on polar equal area projection. Symbols as in Fig. 3.

reverse, three normal) after a.f. cleaning in 200 Oe for all specimens. Mean directions for individual flows as well as mean of the fourteen normal and four reverse flows is shown in Fig. 5. Two top flows near elevations 3300–3400 ft did not yield any stable directions and these were therefore rejected.

Mean value of declination (D), Inclination (I), NRM intensity (J_N), susceptibility (K), precision parameter (k) and Konigsberger ratio Q_n for each flow *vs* elevation, are plotted in Fig. 6. The figure shows a clear record of field reversal from reverse to normal polarity. The values for susceptibility are typical for basaltic rocks. Q_n ratios shows a large variation from 0·2 to approximately 100. Such variations have however been reported by Nagata (1961) and others. There appears to be a general correlation between NRM intensity and Q_n ratio. It is interesting to note that although most of the sites show Q_n ratio less than one precision parameter (k) (Fisher 1953) for these is quite good (mean value 122)-.

4. Results and discussion

The results of magnetically cleaned directions are given in Table 1. These include mean directions as well as the corresponding Virtual Geomagnetic Poles (VGP), along with other palaeomagnetic parameters. An important part of the result is the record of field reversal from reverse to normal polarity at an elevation between 360 to 470 ft. This field reversal (from reverse to normal) has been recorded in many parts of the Deccan traps as shown in Fig. 7 although the elevation at which it occurs in found to be different in different areas. It may be reported here that there appears to be only one well established major field reversal observed so far in the Deccan traps and this is in the same sense as mentioned above. This may be on account of

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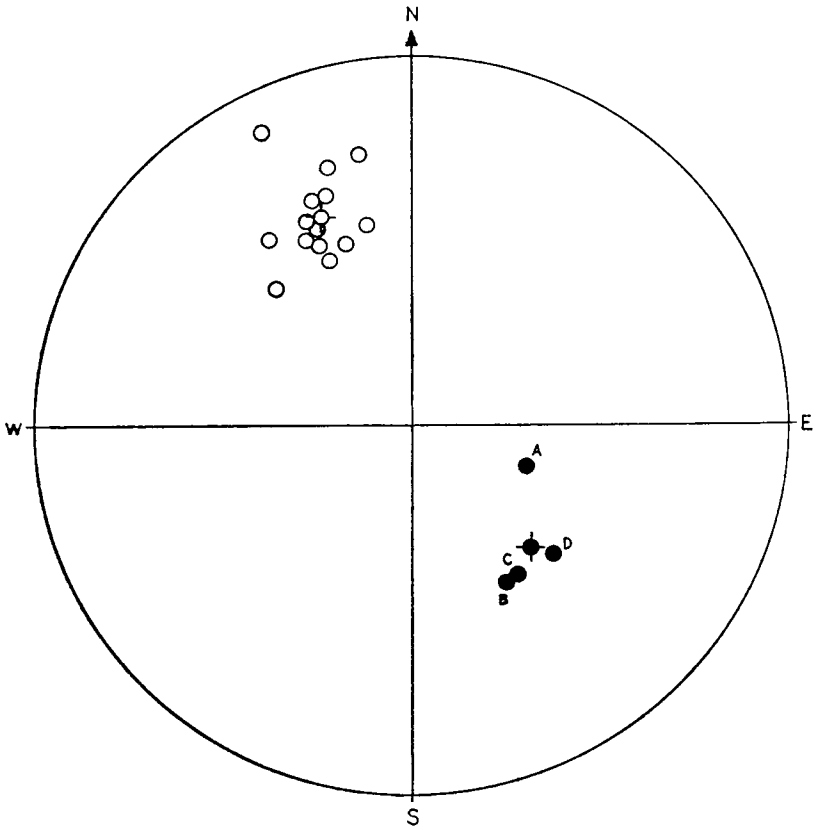


FIG. 5. Mean directions of four flows with reverse polarity and 14 with normal polarity plotted on polar equal area projection. ⊹ mean direction, normal flows. ✦ mean direction reverse flows. ●-down dip, ○-up dip.

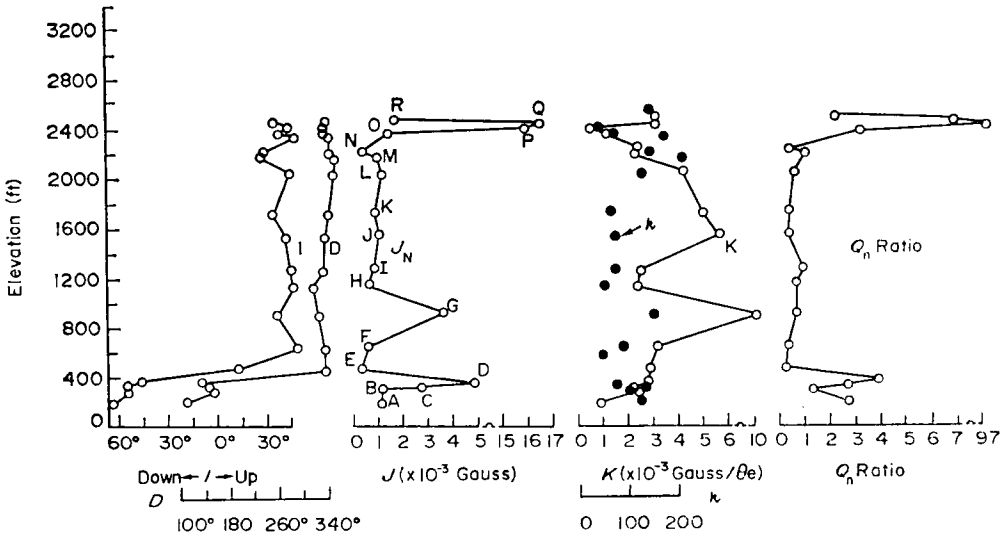


FIG. 6. Mean directions of magnetization of each flow plotted *vs* elevation along with NRM intensity (J_n), susceptibility (K), precision parameter (k) and Koningsberger ratio Q_n.

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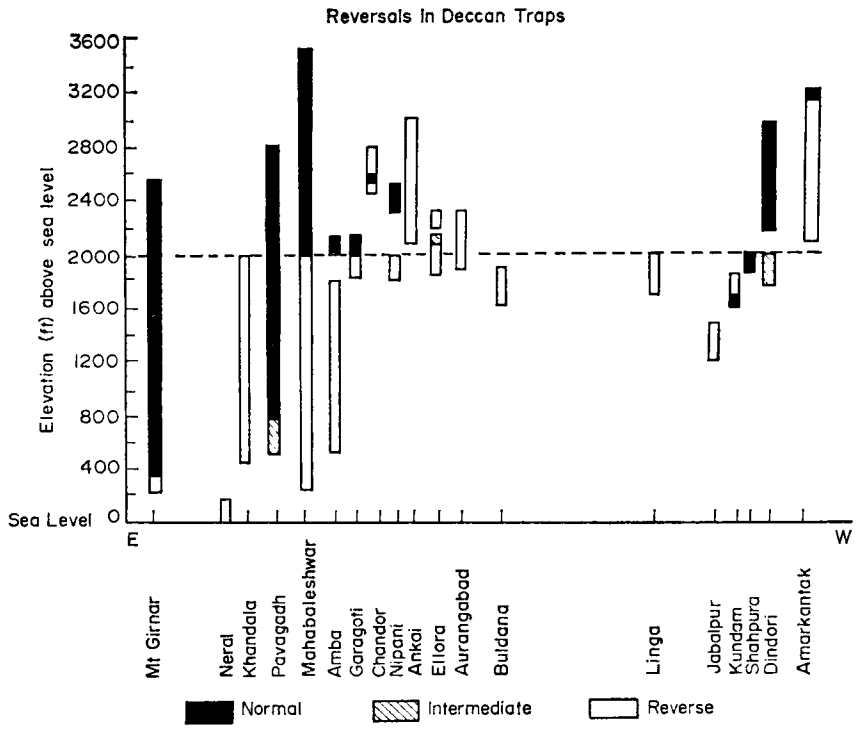


FIG. 7. This shows the nature of polarity reversal observed in various places spread over the territory of the Deccan traps.

lack of detailed sampling or more probably on account of long periods during which field continued to have only one polarity. However very short periods during which the polarity of the field might have changed from normal to reverse as suggested by results from Chandor (Bhamshankaram & Pal 1968) cannot be ruled out.

From all available evidence it appears that the long, period of normal polarity as observed in Mt Girnar is the same as that of Mt Pavagarh (Deutsch *et al.* 1959) or Mahableshwar (Sahsrabudhe 1963) and Jabalpur (Verma & Pullaiah 1972).

5. Palaeosecular variation

Considering that each mean direction of a flow represents a spot reading of geomagnetic in the past we can get some idea about the nature of the field and its secular variation from the scatter of observed flow directions or that of their corresponding Virtual Geometric Poles (Cox & Doell 1964). We have adopted the latter procedure for the present study. The Virtual Geomagnetic Poles corresponding to each flow direction are given in Table 1 and plotted on stereographic projection in Fig. 8 along with their mean. The total angular dispersion (S_T) of Poles observed consists of two parts, between site dispersion (S_B) due mainly to ancient geomagnetic secular variation, and within site dispersion S_w. The two are related as follows (Doell 1970),

$$S_T^2 = S_B^2 + S_W^2/\bar{N},$$

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Table 1

Palaeomagnetic data for 18 flows from Mount Girnar.

Site No.	N	n	Elevation (ft)	0	L_0	D	I	R	k	S.D. (S_W)	α_{95}	Pole position λ_p	L_p(°W)	Polarity
GA	6	26	200	21° 30′ N	70° 30′ E	111	+64	5·96	125	7·2	5° 7′	3° 7′ S	68° 45′	Reverse
GB	6	24	300	21° 30′ N	21° 30′ N	150	+49	5·95	100	8·1	5° 43′	31° 03′ N	79° 06′	Reverse
GC	5	23	310	21° 30′ N	70° 30′ E	146	+49	4·97	133	7·0	5° 26′	29° 5′ N	75° 49′	Reverse
GD	5	21	360	21° 30′ N	70° 30′ N	134	+47	4·95	80	9·0	7°	23° 21′ N	65° 50′	Reverse
GE	6	20	470	21° 30′ N	70° 30′ E	334	−13	5·90	50	11·45	8° 5′	52° 04′ N	64° 23′	Normal
GF	6	27	640	21° 30′ N	70° 30′ E	333	−50	5·94	83	8·90	6° 16′	31° 38′ N	82° 15′	Normal
GG	7	27	990	21° 30′ N	70° 30′ E	323	−37	6·96	150	6·61	4° 19′	34° 28′ N	66° 25′	Normal
GH	6	28	1130	21° 30′ N	70° 30′ E	315	−47	5·90	50	11·45	8° 5′	24° N	66° 30′	Normal
GI	7	34	1260	21° 30′ N	70° 30′ E	333	−45	6·92	75	9·42	6° 7′	35° 17′	79° 40′	Normal
GJ	6	22	1530	21° 30′ N	70° 30′ E	335	−42	5·93	71	9·62	6° 48′	38° 12′ N	80° 08′	Normal
GK	7	19	1720	21° 30′ N	70° 30′ E	340	−35	6·90	60	10·46	6° 50′	44° 45′ N	83° 39′	Normal
GL	7	24	2040	21° 30′ N	70° 30′ E	349	−44	6·95	120	7·4	4° 50′	41° 32′ N	96° 14′	Normal
GM	5	22	2170	21° 30′ N	70° 30′ E	350	−27	4·98	200	5·7	4° 26′	52° 54′ N	93° 18′	Normal
GN	5	22	2210	21° 30′ N	70° 30′ E	343	−29	4·97	133	7·0	5° 26′	49° 25′ N	84° 23′	Normal
GO	6	25	2350	21° 30′ N	70° 30′ E	339	−47	5·97	167	6·37	4° 25′	36° 13′ N	72° 39′	Normal
GP	7	21	2380	21° 30′ N	70° 30′ E	334	−39	6·91	67	9·9	6° 28′	39° 37′ N	77° 39′	Normal
GQ	6	27	2420	21° 30′ N	70° 30′ E	332	−43	5·85	33	14·1	9° 57′	35° 38′ N	77° 15′	Normal
GR	5	18	2470	21° 30′ N	70° 30′ E	337	−35	4·97	133	7·0	5° 26′	43° 24′ N	79°	Normal
Mean (GE–GR)	6·14–		200–2470	21° 30′ N	70° 30′ E	336	38° ½	13·7	43	8·95	5°	40° 53′ N	79° 30′ W	N

N = Total number of samples.
n = Total number of specimens.
λ_0 = Present Latitude of the site.
L_0 = Present Longitude of the site.
D = Mean Declination of stable remanence, East of North.
I = Mean Inclination of stable remanence, Down +ve and up −ve.
R = Length of vector sum of N unit vectors.
k = Precision parameter (Fisher 1953).
α_{95} = Radius of circle of confidence at 95 per cent probability level.
λ_p = Latitude of vertual geomagnetic pole (North Pole).
L_p = Longitude of virtual geomagnetic pole (North Pole).
S.D. = Circular standard deviation.

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where $\bar{N}$ is the average no. of samples per flow. $S_w -$ is taken to be mean of individual flow angular standard deviations give in Table 1 and

$$S_T = (N-1)^{-\frac{1}{2}} \left(\sum_{i=1}^{N} \delta_i^2 \right)^{\frac{1}{2}},$$

where δ_i is the angular deviation of ith pole from the mean pole for the whole sequence. In the present analysis $S_T = 12{\cdot}2°$, $S_w = 8{\cdot}9°$, $S_B = 11{\cdot}7°$ and $N = 6{\cdot}14$. The between site dispersion also usually includes a small component due to local magnetic anomalies that existed at the sampling site at the time of cooling of the lavas. On account of several unknown factors regarding history of eruption of traps and pre-Deccan trap topography no correction has been made for this factor. This is however estimated to be very small, of the order of 1° or less. Ninety-five per cent confidence limits for the standard deviation were calculated according to the method suggested by Cox (1969b). The standard deviation for the normal sequence of flows observed in Mount Girnar is given in Fig. 9 along with similar data from Hawaii, Gallapagos, Western U.S., New Zealand (Cox 1969a), France and Alaska (Doell 1970). In Fig. 9 angular dispersion due to dipole wobble alone and that due to model C of Doell (1970) which combines dipole wobble of 11° with non-dipole field distribution similar to the present one is also plotted. In general palaeosecular variation as shown by the dispersion of Virtual Geomagnetic Poles may be explained as due to either the wobble of the dipole field alone or a combination of the dipole field wobble plus variation of the non-dipole field in strength and direction with respect to the dipole field. Dispersion of VGP's produced by the dipole wobble is invariant with latitude, while randomly directed non-dipole field produces dispersion which varies with respect to latitude (Cox 1970). The angular dispersion for the normal field as observed from Mount Girnar can be explained as being mostly due to the dipole wobble plus perhaps a small contribution from the non-dipole field. It may be remarked here that this dispersion is of the same order of magnitude as that of western United States (Cox & Doell 1964).

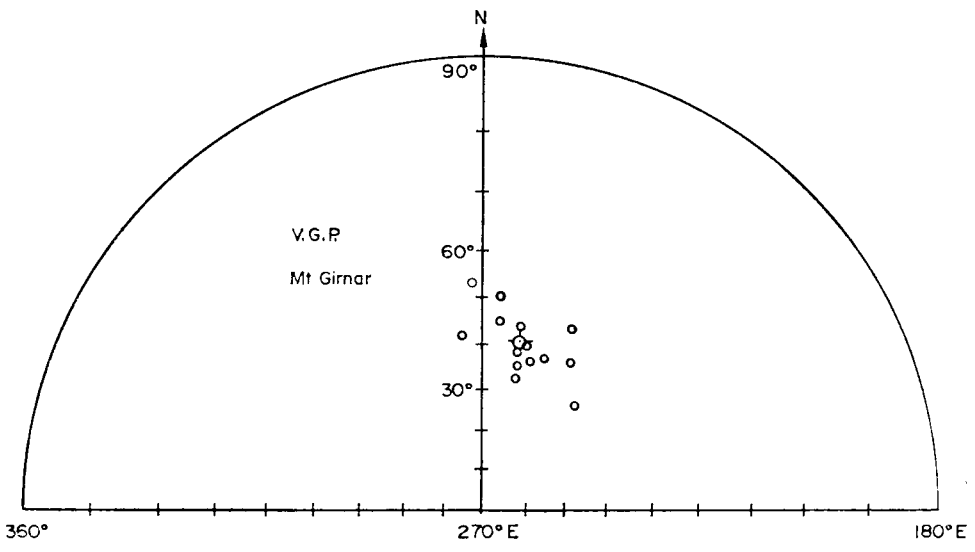


FIG. 8. Virtual Geomagnetic Poles corresponding to each flow with normal polarity plotted on stereographic projection. Mean pole indicated by ⊹.

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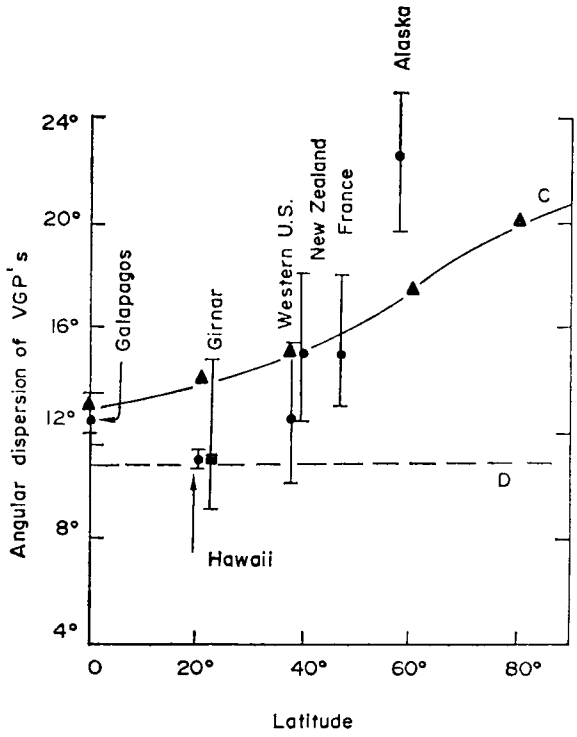


FIG. 9. Angular dispersion of V.G.P.s for the normal sequence of flows observed in Mount Girnar plotted *vs* latitude of sites along with similar data from Hawaii, Gallapagos, Western U.S., New Zealand, France and Alaska (Doell 1970).

6. Continental drift

Various investigators have made estimates of continental drift of the Indian landmass during the period of eruption of the Deccan traps. Deutsch *et al.* (1959) gave an estimate of drift of about 50° in latitude during the past 70 My. This was based upon palaeomagnetic studies of several flows from the western ghats and a few from Mount Pavagarh. McElhinny (1968) on the other hand has suggested that only a few degrees of drift has taken place during the period of entire Deccan trap eruptions which could have lasted for a period of about 5 My.

We have obtained an estimate of the drift of Indian land-mass during the Deccan trap period by studying the mean pole position obtained for several vertical sequences of traps which have been studied in detail from one locality such as Khandala (reverse polarity), Mahableshwar also reverse (Sahsrabudhe 1963), Jabalpur—normal (Verma & Pullaiah 1972) and Mount Girnar—normal (present study). The data for mean directions from these flows, their elevations, pole positions, angular dispersions and other parameters are given in Table 2. The pole positions given in Table 2 are north magnetic poles for the sequences with normal polarity and south poles for sequences with reverse polarity. Pole positions for these sequence are plotted in Fig. 10 along with limits of their probability error at 95 per cent confidence level.

The exact time span represented by the igneous activity in these areas is difficult to estimate. However if the average time between successive flows is of the order of 1000 years as estimated by Doell & Cox (1965) for Hawalin Islands, the secular variation should be averaged out in each of the sequences studied. On this assumption we obtain a drift of the order of 23° in latitude between the time of eruption of

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Table 2

Palaeomagnetic parameters of vertical sequences of flows from five localities in the Deccan traps.

No.	Locality	Elevation (ft)	No. of flows (N)	Mean direction D	I	R	k	α_{95}	λ_0(°S)	Virtual Geomagnetic Pole λ_p(°N)	L_p(°W)	dp	dm	Polarity	Reference
1	Jabalpur 23° 04′ N, 80° 57′ E	2150–2900	8 flows	343	−28	7·92	87	5	15	48	74	3	6	N	Verma & Pullaiah 1972
2	Mt. Girnar 21° 30′ N, 79° 30′ E	440–2470	14 flows	336	−38½°	13·7	43	4	21° 40′	41	79½	4	7	N	Present work
3	Mahableshwar 17° 55′ N, 73° 38′ E	200–2000	15 flows	153	+52	13·80	12	11	33	33	33	80	10	R	Sahsrabudhe (1963)
4	Khandala 18° 45′ N, 73° 22′ E	400–2000	15 flows	147	+58	14·70	47	5	39	25	79	5	7	R	Sahsrabudhe (1963)

D = declination in degrees clockwise from geographic north, and I = inclination in degrees, +ve down and −ve up.
R = Resultant of N unit vectors, and $k(= N-1/N-R)$ Fisher's precision parameter.
95 = Circle of confidence at 95 per cent probability level.
λ_0 = ancient latitude of the site.
λ_p and L_p = latitude and longitude, respectively, of the virtual geomagnetic pole (VGP)
dp and dm = semi-axes of the elliptical error area round the pole at 95 per cent probability level.
N = normal polarity, and R = reverse polarity.

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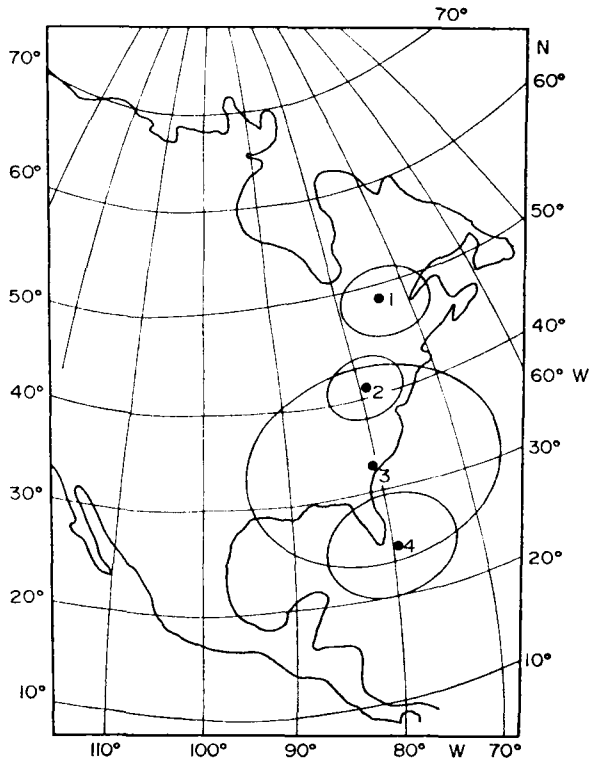


FIG. 10. Pole positions corresponding to mean pole for four vertical sequences of traps from Jabalpur, Mt Girnar, Mahableshwar, and Khandala are shown along with ovals of confidence at 95 per cent probability level. Data for each sequence are given in Table 2. 1, Jabalpur; 2, Girnar; 3, Mahabaleshwar; 4, Khandala.

Khandala flows and those of Jabalpur area. Although the error in pole position No. 3, for Mahableshwar is very large, those for poles 1, 2 and 4 are within reasonable limits and the poles are significantly different from each other. The normal flows from Mount Girnar give a pole position which is intermediate between Jabalpur (N) and Mahableshwar (R) suggesting that the igneous activity represented by flows from Mt Girnar is of intermediate age between those of Mahableshwar and Jabalpur.

Considering that the age of the Deccan traps from Mount Girnar is around 64 My as given by McElhinny (*ibid*) our data suggest that the flows from Khandala area are much older and those from Jabalpur area are relatively younger. No radioactive ages for flows from these two areas have yet been obtained. It is highly desirable to do so.

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